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# The impact of prior chatbot identity disclosure on customer tolerance of service failures

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## Abstract

**Purpose** – Large language models, such as ChatGPT, have made chatbot responses seem more natural, making it more difficult for customers to distinguish that chatbots are non-human. Although chatbots offer advantages in the service industry, their technical limitations can lead to service failures. This study aims to explore how the disclosure of the chatbot's identity during the initial stage of service, before any service failure occurs, affects customer tolerance of service failures.

**Design/methodology/approach** – Three psychological experiments were conducted to assess the effects of chatbot identity disclosure on customer tolerance of service failure.

**Findings** – This study confirms that the disclosure of the chatbots' identity at the initial stage of service, before any service failure occurs, has a positive impact on consumers' tolerance of service failures. Compared to non-identity disclosure, chatbot identity disclosure increases consumers' perception of chatbots' sincerity and reduces their disgust, thus increasing the consumers' tolerance of chatbot service failures. In addition, the explanation of service failures moderates the relationship between chatbot identity disclosure and customers' feelings of disgust and perceived sincerity.

**Practical implications** – This study reveals the importance of chatbot identity disclosure for service companies. Managers should consider a chatbot's ability to handle service failures when deciding whether to disclose its identity.

**Originality/value** – This research contributes to the literature on the positive effects of identity disclosure by investigating how the disclosure of a chatbot's identity during the initial stage of service, prior to any service failure, affects consumers' tolerance of service failures.

**Keywords** Chatbot identity disclosure, Service failure and recovery, Customer tolerance of service failure, Disgust, Perceived sincerity

**Paper type** Research article

## 1. Introduction

In 2024, *Research and Markets* reported that the global chatbot market had attained a value of $7.01 billion and was projected to reach $20.81 billion by 2029 (Mordor Intelligence, 2024). Despite the widespread application of chatbots in online customer service (Mozafari *et al.*, 2022), customer acceptance of this technology is not uniform. Prevalent biases and resistance constitute major obstacles to the widespread adoption of chatbots in the service industry. In particular, consumers harbor doubts about whether chatbots can provide effective service. When there is no significant difference in service performance, customers tend to rate chatbots more favorably when the chatbots do not disclose their bot identity, compared to when they explicitly do (Castelo *et al.*, 2023; Luo *et al.*, 2019). This preference indicates that the disclosure of a chatbot's identity may influence consumer trust and evaluation, which in turn



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affect business performance. Consequently, given technological advancements that allow chatbots to convincingly mimic human language, many companies choose not to actively disclose the non-human identity of their chatbots (Li *et al.*, 2024). Nonetheless, chatbots will inevitably make some mistakes during service delivery; examples include inaccurate semantic understanding or inappropriate responses (Xie *et al.*, 2024). In such situations of service failure, if a chatbot does not disclose its true identity, consumers may deem the chatbot to be untrustworthy, and this can negatively affect corporate performance (Mozafari *et al.*, 2022). Therefore, the question of whether to disclose the identity of a chatbot in the context of service failure has become a dilemma for the modern service industry.

Currently, studies to date have primarily focused on exploring the potential negative effects of identity disclosure (De Cicco *et al.*, 2021; Li *et al.*, 2024; Luo *et al.*, 2019), whereas the potential positive impacts of identity disclosure remain underexplored. Studies that specifically examine the effects of identity disclosure in the context of service failure are particularly rare. Specifically, most previous studies have suggested that identity disclosure reduces social presence and empathy and thereby negatively influences customers' purchasing behavior and their attitudes toward retailers (De Cicco *et al.*, 2021; Luo *et al.*, 2019). To reduce the adverse effects of identity disclosure, scholars have proposed various strategies, such as delaying identity disclosure or providing customers with information on chatbot expertise and limitations (Luo *et al.*, 2019; Mozafari *et al.*, 2021). However, some studies have suggested that the disclosure of chatbots' identity actually increases users' perceived closeness and trust in the chatbot (Kim, 2025). In the context of service failure, Mozafari *et al.* (2022) found that identity disclosure by chatbots can enhance customer trust and loyalty. However, Funke *et al.* (2023) were unable to confirm this result in other industry sectors. Moreover, the above studies are all based on identity disclosure at the conclusion of the service phase. It is worth noting that disclosure of the presence of other customer service representatives in the late stage of service failure may trigger emotional responses in customers, thereby stimulating the customers' aggressive behavior (Huang and Dootson, 2022). Thus, the aim of this study is to explore the impact of the disclosure (vs the non-disclosure) of chatbots' identity on consumer responses to service failure at the initial stage of interaction, before service failure occurs.

To address the shortcomings of existing studies, this study focuses on the initial stage of service delivery to explore the impact of chatbot identity disclosure on consumer tolerance of service failure, as well as its potential mechanisms. Two mediating pathways are proposed to explain consumer reactions to chatbot identity disclosure. Specifically, working from the framework of social exchange theory, this study posits that the disclosure of chatbots' identity can enhance consumers' perceptions of the chatbots' sincerity and reduce their feelings of disgust toward chatbots, thereby increasing their tolerance of service failures. Moreover, research on explainable artificial intelligence (AI) has shown that explanations allow users to comprehend the decision-making process of AI (Brasse *et al.*, 2023). Providing timely explanations after a service failure has been demonstrated to help repair consumers' social evaluations of robots (Choi *et al.*, 2021). Consequently, this paper suggests that if consumers receive accurate and transparent information about service failures, it may impact their responses to the initial disclosure or non-disclosure of chatbots' identity.

This research fills the above-mentioned knowledge gaps and makes multiple contributions to the study of chatbots in the service industry. First, prior studies have paid little attention to the positive effects of chatbot identity disclosure, particularly in the context of service failure. Both Mozafari (2022) and Funke *et al.* (2023) focused on service outcomes as moderators and on late-stage identity disclosure after service failure. In contrast, this study explores how identity disclosure at the initial stage of interaction, before service failure, affects consumers' tolerance of service failure. This is a useful addition to the existing body of research on identity disclosure. Second, this study expands the understanding of customer tolerance of chatbots' service failure by introducing the mediating effects of disgust and perceived sincerity in the influence of chatbot identity disclosure. This novel perspective contributes to the application of social exchange theory in the context of chatbot service. Third, this study identifies effective boundary conditions

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for the effects of identity disclosure by examining the moderating role of explanation. Therefore, this research contributes to the literature on AI transparency and explainable AI.

## 2. Literature review

### *2.1 Chatbots in service failure and recovery*

Chatbots are software agents that can interact and communicate with consumers through text or voice (Brandtzaeg and Følstad, 2018). Currently, they have been extensively deployed across service domains such as hospitality and online retail. Chatbots' ability to understand natural language and conduct conversations makes it possible to use them in place of humans as service providers (McLean and Osei-Frimpong, 2019). However, the currently inadequate ability of AI to achieve empathetic intelligence (Huang and Rust, 2018) means that service failures are inevitable.

Service recovery consists of a series of responsive actions taken by service providers to alleviate consumer dissatisfaction and complaints following service failures. Such actions typically involve compensation, response speed, apologies, consumer choice and restoration initiation (Schütte *et al.*, 2017). The aim is to rectify errors and attempt to evoke forgiveness from consumers, thereby ensuring their satisfaction. Prior research on service recovery by intelligent service agents has primarily focused on appearance design, verbal recovery, language style and human employee intervention. First, the high anthropomorphism (vs low anthropomorphism) of intelligent objects has been shown to effectively reduce consumer dissatisfaction following service failures (Choi *et al.*, 2021; Fan *et al.*, 2020) and to influence intention conversion (Lu *et al.*, 2024). For instance, the perceived warmth (vs perceived competence) of robots' appearance plays a significant role (Chang *et al.*, 2024). Second, expressing appreciation for consumers' correction of service failures (vs apologies) has been shown to increase consumers' concerns about "face", thereby enhancing their satisfaction after recovery (Song *et al.*, 2023). Third, expressing empathy via language can effectively increase trust by reducing psychological distance, thereby enhancing customers' willingness to engage in continuous usage (Lv *et al.*, 2022). In addition, the use of cute and/or humorous language has been shown to improve consumers' tolerance of chatbots (Lv *et al.*, 2021; Xu and Liu, 2022). Fourth, studies have shown that apologies delivered by human employees are perceived as more natural and sincere, compared to those from chatbots. Thus, human apologies are better for enhancing customer satisfaction (Zhang *et al.*, 2023; Zhu *et al.*, 2023).

Given that human intervention is not immediate, and responding often requires time, the ability of chatbots to self-recover is crucial for alleviating consumers' negative emotions. However, chatbots have a limited ability to accurately assess customer losses, demonstrate empathy and provide appropriate emotional support (Huang and Rust, 2018; Lv *et al.*, 2021), and thus their available strategies for service recovery are relatively limited. Some studies have focused on addressing service failures through chatbot appearance design, verbal recovery and language style, but there have been too few studies on chatbot identity disclosure strategies. Chatbots are commonly perceived as lacking emotional intelligence (Zhang *et al.*, 2023). Therefore, when consumers realize that they are dealing with a chatbot instead of a human, their expectations regarding chatbot performance will be relatively low (Grimes *et al.*, 2021). The expectation disconfirmation model posits that, when actual experiences fall short of expectations, dissatisfaction will be lower under conditions of low expectations rather than high (Gursoy *et al.*, 2019). Hence, it is necessary to explore whether disclosing the non-human identity of chatbots before service failures occur can reduce consumer dissatisfaction and thereby increase their tolerance of service failures.

### *2.2 Chatbot identity disclosure*

Disclosure is the practice of explicitly informing customers that they are communicating with a chatbot rather than a human employee, before service is delivered (Luo *et al.*, 2019). For

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example, in some online stores, customer service chatbots disclose their identity at the beginning of the interaction with customers, using sentences like, "Dear customer, I am the chatbot of this store." (Cheng *et al.*, 2022). Chatbot identity disclosure has been recognized as a key factor affecting customers' responses to chatbot encounters (Li *et al.*, 2024; Luo *et al.*, 2019; Mozafari *et al.*, 2021) (Table 1).

Prior studies that have focused on identity disclosure argue that the disclosure of non-human identity will impede consumers' responses to chatbots (De Cicco *et al.*, 2021; Li *et al.*, 2024; Luo *et al.*, 2019; Powers *et al.*, 2023). Compared to undisclosed chatbots, consumers perceive chatbots that disclose their identity as lacking knowledge, sympathy or any sense of social presence, and this decreases the consumers' willingness to purchase (De Cicco *et al.*, 2021; Luo *et al.*, 2019). In videos recommending stock trading, disclosing avatars as AI entities has similarly been shown to reduce customer trust and purchasing behavior (Powers *et al.*, 2023). To mitigate these negative impacts of identity disclosure, scholars have proposed both optimizing the timing of disclosure (Luo *et al.*, 2019; Zhu and Zhang, 2024) and enhancing disclosure content (Meng *et al.*, 2023; Mozafari *et al.*, 2021).

However, other studies have suggested that revealing a chatbot's identity can have positive effects, such as enhancing users' sense of intimacy and trust in the robot (Kim, 2025). For instance, Beeler *et al.* (2022) found that disclosing the use of a digital assistant before (vs after) interaction can increase (vs decrease) the perceived competence of the digital assistant, resulting in better consumer outcomes. In cases of service failure, disclosing the identity of the chatbot to consumers can have a positive impact on their trust and on customer retention (Mozafari *et al.*, 2022). Moreover, some studies have indicated that the effects of identity disclosure depend on customers' anxiety. For example, identity disclosure increases perceived certainty in anxious consumers, thereby enhancing their satisfaction and purchase intention. However, for non-anxious customers, the effect of identity disclosure is not significant (Yang and Shi, 2022).

To date, there has been relatively little discussion in literature regarding service failures, and the academic community has not reached a consensus on this issue. Mozafari *et al.* (2022) suggested that, in the event of a service failure, the disclosure of a chatbot's identity to customers may help improve customer retention. However, Funke *et al.* (2023) found that, regardless of whether the service provided by a chatbot was successful or not, the disclosure of the chatbot's identity had no significant effect on customer retention. It is worth noting that in both studies, the identity of the chatbot was disclosed at the end of the interaction after the service failure, and this timing may have led to negative customer reactions. According to Huang and Dootson (2022), revealing too late the involvement of human employees after a service failure may prompt customers to handle issues emotionally, leading to aggressive behavior. Currently, no studies have explored the effects of disclosing the identity of a chatbot at the beginning of an interaction, before a service failure occurs, on consumer reactions to those service failures. Given that identity disclosure is known to enhance the positive relationship between chatbots' friendliness and customer trust (Cheng *et al.*, 2022), this study aims to investigate the impact of disclosure of chatbot identity at the initial stage of service on customer tolerance of service failure.

## 3. Hypotheses development

### *3.1 Chatbot identity disclosure and customer tolerance of service failure*

As described by expectancy violations theory, when working with chatbots, consumers use cues to form expectations about future interactions. The identity cue influences expectations regarding the service offered by chatbots (Grimes *et al.*, 2021). The chatbot's disclosure of its identity reinforces consumers' belief that the AI lacks the ability to handle uncertainty (Kim and Duhachek, 2020), leading them to lower their expectations of the chatbot. Alternatively, consumers tend to regard a non-disclosed chatbot as being a real, social human, engendering high expectations for interaction performance (Grimes *et al.*, 2021). The occurrence of service

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**Table 1.** Review of the literature on chatbot identity disclosure

| Studies | Service outcomes | Disclosure types | Disclosure stage | Disclosure information | Independent variables | Dependent variables | Mediators | Moderators | Key findings |
|---|---|---|---|---|---|---|---|---|---|
| Go and Sundar (2019) | No service failure | Chatbot | Pre-service stage | Identity | 2 (Anthropomorphic visual cues: High vs Low) × 2 (Message interactivity: High vs Low) × 2 (Identity cue: Chatbot vs Human) | Attitudes, behavioral intentions | Social presence, perceived homophily | – | Chatbots that claim human identity, compared to those that disclose their identity as robots, are expected to exhibit higher social presence and perceived homophily |
| Luo *et al.* (2019) | No service failure | Chatbot | Pre-service stage | Identity | Undisclosed chatbots vs Disclosed chatbots vs Human | Customer purchases | Knowledgeable, empathetic | Late disclosure timing strategy, customers' prior AI experience | Disclosed chatbots, compared to undisclosed ones and human employees, are perceived as being less knowledgeable and empathetic, resulting in lower customer purchase rates |
| De Cicco *et al.* (2021) | No service failure | Chatbot | Pre-service stage | Identity | Undisclosed chatbots vs Disclosed chatbots | Attitude toward the online retailer | Social presence, trust | – | Compared with undisclosed chatbots, disclosed chatbots evoke lower levels of social presence and trust, which in turn negatively affect attitudes toward online retailers |

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**Table 1.** Continued

| Studies | Service outcomes | Disclosure types | Disclosure stage | Disclosure information | Independent variables | Dependent variables | Mediators | Moderators | Key findings |
|---|---|---|---|---|---|---|---|---|---|
| Grimes *et al.* (2021) | No service failure | Chatbot | Pre-service stage | Identity | Conversational capability | Perceived conversational engagement | – | Expectation violations | Chatbots that disclose their robot identity, compared to those that claim human identity, face lower consumer expectations of their conversational abilities |
| Mozafari *et al.* (2021) | No service failure | Chatbot | Pre-service stage | Content | No identity disclosure vs Mere identity disclosure vs Disclosure with communication of expertise vs Disclosure with communication of weaknesses | Trust in the conversational partner | – | – | Compared to non-disclosure, disclosure of a chatbot's identity reduces trust, but providing information about expertise or weaknesses mitigates this effect |

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**Table 1.** Continued

| Studies | Service outcomes | Disclosure types | Disclosure stage | Disclosure information | Independent variables | Dependent variables | Mediators | Moderators | Key findings |
|---|---|---|---|---|---|---|---|---|---|
| Beeler *et al.* (2022) | No service failure | Chatbot | Pre-service stage | Identity | Study 1: Digital assistant use contexts (Automation vs Augmentation)<br>Study 2: Disclosure (Yes vs No) | Consumer outcomes | Ability | Study 1: Consumer preference for human interaction,<br>Study 2: Consumer positive mood | Disclosing the use of digital assistants before interaction increases the perceived abilities of the digital assistant, resulting in higher consumption outcomes. However, disclosing after interaction decreases perceived abilities, leading to lower consumption outcomes |

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**Table 1.** Continued

| Studies | Service outcomes | Disclosure types | Disclosure stage | Disclosure information | Independent variables | Dependent variables | Mediators | Moderators | Key findings |
|---|---|---|---|---|---|---|---|---|---|
| Cheng *et al.* (2022) | No service failure | Chatbot | Pre-service stage | Identity | Empathy, friendliness of the chatbot | AI reliance, AI resistance | Trust | Task complexity, chatbot identity disclosure | While the disclosure of chatbot identity weakens the positive impact of perceived empathy on trust, the disclosure strengthens the positive impact of perceived friendliness on trust, resulting in higher reliance and lower resistance to chatbots |
| Huang and Dootson (2022) | Service failure | Human | – | Timing | The timing of disclosing the presence of human employees | Customer aggression | Emotion-focused coping, problem-focused coping | Customer participation | Compared to early disclosure, the late disclosure of the availability of human employees is more likely to heighten emotion-focused coping and trigger customer aggression |

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**Table 1.** Continued

| Studies | Service outcomes | Disclosure types | Disclosure stage | Disclosure information | Independent variables | Dependent variables | Mediators | Moderators | Key findings |
|---|---|---|---|---|---|---|---|---|---|
| Mozafari *et al.* (2022) | – | Chatbot | Post-service stage | Identity | Study 1: Service criticality (High vs Low service criticality) × Disclosure (Undisclosed chatbots vs Disclosed chatbots) Study 2: Service outcome (Chatbot failure vs No chatbot failure) × Disclosure (Undisclosed chatbots vs Disclosed chatbots) | Customer retention | Trust | Study 1: Service criticality (High vs Low), Study 2: Service outcome (Chatbot failure vs No chatbot failure) | In situations where service criticality is low or service failures occur, chatbot identity disclosure can have a positive impact on trust, thereby increasing customer retention |
| Yang and Shi (2022) | No service failure | Chatbot | Pre-service stage | Identity | Customer anxiety (Yes vs No) × Disclosure (Undisclosed chatbots vs Disclosed chatbots) | Customer satisfaction, company evaluation, purchase intention | Perceived certainty, pre-interaction expectation for the chatbot | Customer anxiety (Yes vs No) | For anxious customers, identity disclosure mitigates the negative impact of customer anxiety on customer satisfaction, company evaluation and purchase intention. However, for non-anxious customers, this mitigating effect is not observed |

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**Table 1.** Continued

| Studies | Service outcomes | Disclosure types | Disclosure stage | Disclosure information | Independent variables | Dependent variables | Mediators | Moderators | Key findings |
|---|---|---|---|---|---|---|---|---|---|
| Castelo *et al.* (2023) | No service failure | Chatbot | Pre-service stage | Identity | Human vs Service bot | Overall service evaluation | Cost-cutting attributions | – | Under the same service outcomes, chatbots that claim human identity receive higher service evaluations than those that disclose their identity as robots. However, consumers only give chatbots that disclose their identity as robots good service evaluations when their effectiveness surpasses that of chatbots that claim human identity |

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**Table 1.** Continued

| Studies | Service outcomes | Disclosure types | Disclosure stage | Disclosure information | Independent variables | Dependent variables | Mediators | Moderators | Key findings |
|---|---|---|---|---|---|---|---|---|---|
| Funke *et al.* (2023) | – | Chatbot | Post-service stage | Identity | Service outcome (Chatbot failure vs No chatbot failure) × Disclosure (Undisclosed chatbots vs Disclosed chatbots) | Customer retention | Trust | – | Regardless of the success or failure of chatbot services, the disclosure of chatbots has no impact on customer retention and service outcomes. Only service outcomes have an impact on customer retention |
| Kirkby *et al.* (2023) | No service failure | AI | – | Identity | AI vs Human | Brand attitude | Brand voice authenticity, brand authenticity | Emotionality | AI-generated text is not considered less authentic than text presented as having been written by humans, nor does it negatively affect the perceived authenticity of brand voice or brand attitude |

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**Table 1.** Continued

| Studies | Service outcomes | Disclosure types | Disclosure stage | Disclosure information | Independent variables | Dependent variables | Mediators | Moderators | Key findings |
|---|---|---|---|---|---|---|---|---|---|
| Meng *et al.* (2023) | No service failure | Chatbot | Pre-service stage | Content | One-sided message vs Double-sided message | Customers' willingness to communicate with AI chatbots | Perceived authenticity | Customer demand types | Compared with one-sided messaging strategies, two-sided messaging strategies are more likely to improve the perceived authenticity for customers, thereby increasing their willingness to communicate with chatbots |
| Powers *et al.* (2023) | No service failure | AI | Post-service stage | Identity | Disclosure of an AI-synthesized human avatar vs Non-disclosure | Purchase intention | Perceptions of source trustworthiness, expertise, physical attractiveness | Social cynicism | In the video, disclosure (compared to non-disclosure) of the identity of the AI-synthesized human avatar reduces trust, perceived expertise and physical attractiveness, thereby lowering purchase intention |

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**Table 1.** Continued

| Studies | Service outcomes | Disclosure types | Disclosure stage | Disclosure information | Independent variables | Dependent variables | Mediators | Moderators | Key findings |
|---|---|---|---|---|---|---|---|---|---|
| Gnewuch *et al.* (2024) | No service failure | Human | – | Timing | Up-front human involvement disclosure vs No up-front disclosure, Step-in human involvement disclosure vs No step-in disclosure | Customer communication style, employee workload | Impression management concerns | – | Disclosing human involvement before or during interaction with hybrid service agents increases customer concerns about impression management and leads to a more human-oriented communication style. A more human-oriented communication style ultimately increases employees' workload |
| Li *et al.* (2024) | No service failure | Chatbot | Pre-service stage | Identity | Undisclosed AI agents vs Disclosed AI agents | Consumers' unethical behavior | Perceived social judgment | Consumers' motive for impression management | Compared to undisclosed AI agents, disclosure of AI agents reduces consumers' perceived social judgments, leading to unethical behavior |

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**Table 1.** Continued

| Studies | Service outcomes | Disclosure types | Disclosure stage | Disclosure information | Independent variables | Dependent variables | Mediators | Moderators | Key findings |
|---|---|---|---|---|---|---|---|---|---|
| van der Goot *et al.* (2024) | No service failure | Chatbot | Pre-service stage | Identity | Undisclosed chatbots vs Disclosed chatbots | Mindless and mindful anthropomorphism, social presence, source orientation | – | – | Regardless of chatbot identity disclosure, it is crucial for customer service to demonstrate willingness and a friendly tone to assist users effectively |

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**Table 1.** Continued

| Studies | Service outcomes | Disclosure types | Disclosure stage | Disclosure information | Independent variables | Dependent variables | Mediators | Moderators | Key findings |
|---|---|---|---|---|---|---|---|---|---|
| Xie *et al*. (2024) | No service failure | Chatbot | Pre-service stage | Identity | Humor (Absent vs Present) | Service satisfaction | Perceived competence, entertainment social presence | Chatbot identity disclosure | Identity disclosure by chatbots moderates the relationship between humor and ability/ entertainment/ social presence. When the identity of the chatbot is disclosed, humor has a positive impact on perceived competence, entertainment and social presence, thereby increasing service satisfaction. Conversely, if the identity of the chatbot is not disclosed, this effect is weakened or may be reversed |

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**Table 1.** Continued

| Studies | Service outcomes | Disclosure types | Disclosure stage | Disclosure information | Independent variables | Dependent variables | Mediators | Moderators | Key findings |
|---|---|---|---|---|---|---|---|---|---|
| Zhu and Zhang (2024) | No service failure | Robot chef | – | Timing | Robot chef identity disclosure (After consumption decision vs Before consumption decision) | Consumer perception of ethicality | Feeling of deception | The certainty of food quality before purchase | Revealing the identity of robot chefs after (vs before) consumption decisions increases a sense of deception, reduces consumers' moral perception and thus reduces their willingness to continue using the robot chefs |
| Kim (2025) | No service failure | Chatbot | Pre-service stage | Identity | Identity perception of the bot (Human vs Machine) × Self-disclosure (Personal self-disclosure vs Factual self-disclosure) | Intimate with the robot, trust in the robot | Message interactivity | Users' culture-related trait | When a chatbot is perceived as human and engages in personal self-disclosure, users feel more intimate with and trust the chatbot. When the chatbot is perceived as a machine and engages in factual self-disclosure, users feel more intimate with and trust the chatbot |

**Source(s):** Table by authors

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failure results in a violation of expectations in that the performance is inconsistent with the consumer's expectations. Consumers who have higher expectations of non-disclosed chatbots in the initial service stage will perceive a greater violation of expectations after service failure. An increase in negative interactions will lead to changes in both the evaluation of the chatbot and purchase intention (Crolic *et al.*, 2022; Grimes *et al.*, 2021). Therefore, this study assumes that chatbot identity disclosure (vs non-disclosure) will promote customer tolerance of service failure. The following hypothesis is thus proposed:

*H1*. Chatbot identity disclosure (vs non-disclosure) increases customer tolerance of service failure.

### *3.2 The mediating role of disgust*

Social exchange theory posits that exchange is fundamental to social interactions and that individuals' exchange behaviors are motivated by the rewards they can obtain (Blau, 2017). Thus, individuals are inclined to engage in social exchange activities when they anticipate the potential for immediate or future rewards. Studies have shown that customers tend to perceive chatbots with undisclosed identities as possessing human-like social capabilities. Thus, the consumers are more likely to evaluate their interactions with these chatbots according to the standards of human interpersonal interactions (Luo *et al.*, 2019). Interestingly, compared to chatbots with disclosed identities, consumers hold higher expectations for the conversational abilities of chatbots with undisclosed identities (Grimes *et al.*, 2021). Service failures, which occur when social interactions fail to meet consumer expectations, may lead to conflicts and negative emotions (Burgoon and Hale, 1988). Thus, when consumers have high expectations of their interactions with chatbots of undisclosed identity, service failures can result in more intense disappointment and negative emotions. Disgust is one of the fundamental human emotions (Rozin *et al.*, 1994; Westbrook, 1987). When consumers' high expectations for chatbots with undisclosed identity conflict with actual service failures, they may experience strong feelings of disgust. Therefore, if chatbots with undisclosed (vs disclosed) identity fail to meet consumers' high expectations, that failure may elicit emotions of disgust that consequently affect the customers' tolerance of service failure. Disgust can trigger negative behaviors from customers (Rozin *et al.*, 1994), such as expressing dissatisfaction with the service provider and even terminating the service relationship entirely (Ligas and Coulter, 2001). Consumers who experience disgust with algorithms will often abandon their use of algorithms and develop negative attitudes toward algorithms in general (Dietvorst *et al.*, 2015). The extent of a customer's disgust may impact their level of tolerance of service failures. Thus, the following hypothesis is put forth:

*H2*. Disgust mediates the relationship between chatbot identity disclosure and customer tolerance of service failure, such that non-disclosure (vs disclosure) elicits stronger disgust, thereby reducing customer tolerance of service failure.

### *3.3 The mediating role of perceived sincerity*

Perceived sincerity is an important and widely used factor for service recovery with both human employees and chatbots (Hu *et al.*, 2021; Migacz *et al.*, 2018; Yang *et al.*, 2022). Social exchange theory suggests that interpersonal interactions involve a process of exchanging values (Blau, 2017). Individuals infer interlocutors' sincerity from the content (Rosenblum *et al.*, 2020) and manner of their speech (Hornsey *et al.*, 2020) while communicating. When consumers interact with a chatbot, they infer its sincerity based on the conversation and the chatbot's behavior. For example, in situations where service failures are more severe, the humor (vs rational language) used by a robot may convey signals indicating a lack of seriousness and an insincere attitude (Yang *et al.*, 2022). In addition, when playing rock–paper–scissors with robots, cheating behavior exhibited by robots has been found to

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influence people's perception of the robots' sincerity (Short *et al.*, 2010). In the context of customer service interactions, the practice of chatbots not proactively disclosing the availability of human employees represents a violation of customers' informational rights (Huang and Dootson, 2022). Consumers may believe that non-disclosed chatbots are lying to them out of self-interest and that the chatbots lack sincerity in their interactions. Alternatively, chatbot identity disclosure may be viewed as a sign of sincere service.

Perceived sincerity positively influences customer behaviors. For example, authentic and sincere service providers obtain greater rewards or "service tips" from customers, generating a mutually beneficial outcome for both parties (Pugh, 2001; Tsai, 2001), while also boosting satisfaction and purchase intention (Gountas *et al.*, 2011). Furthermore, when customers perceive a chatbot's responses to be genuine and sincere, the customers are likely to positively evaluate the robot's services (Shan *et al.*, 2024; Yang *et al.*, 2022). By disclosing their identity, chatbots aim to elicit friendly treatment from consumers. When chatbots disclose their true identity to consumers, the consumers perceive them as being more sincere. In the event of service failure, consumers are more tolerant of the failure when they perceive the chatbot as being sincere. Thus, perceived sincerity can help to alleviate consumers' negative reactions to chatbot service failures and improve the consumers' tolerance of such failures. The following hypothesis is therefore proposed:

*H3.* Perceived sincerity mediates the relationship between chatbot identity disclosure and customer tolerance of service failure, in that non-disclosure (vs disclosure) elicits weaker perceptions of sincerity, thereby reducing customer tolerance of service failure.

### *3.4 The moderating role of explanation*

Enhanced explainability not only unlocks chatbots' potential but also effectively mitigates the risks associated with insufficient transparency (Brasse *et al.*, 2023). Following a service failure, the adoption of explanation strategies by chatbots can influence consumers' responses to the disclosure of the chatbot's identity. From the perspective of equity theory, numerous studies have suggested that timely and sincere explanations can compensate consumers by providing psychological and social resources, thus achieving a sense of informational fairness (Choi *et al.*, 2021; Liao, 2007). Furthermore, explanations imply that consumers are provided with reasons for service failures, reducing their concerns and uncertainty about responsibility and thus prompting them to accept and forgive (Zhou and Chang, 2024). Prior studies have indicated that after service failures, highly-anthropomorphized (vs low-anthropomorphized) chatbots can enhance consumer satisfaction through recovery measures, such as providing explanations (Choi *et al.*, 2021). Therefore, this study considers the chatbot's explanation as a moderating factor to explain when chatbot identity disclosure is effective.

When chatbots provide explanations as a means of service recovery, the content of the explanations is aimed at ensuring that consumers have accurate and transparent information about service failures so that they know what actions to take to resolve the issues. Tailoring the recovery process to meet consumers' needs demonstrates attention to detail and a sincere desire to make amends (Zhou and Chang, 2024). At this point, consumers' focus shifts to the content of the explanations, making it easier for the chatbot's identity to be overlooked. In other words, neither the disclosure nor non-disclosure of the chatbot's identity before the interaction will significantly affect consumers' perceptions of the chatbot's sincerity or their disgust. Thus, their tolerance of failure will not be affected. By contrast, when chatbots do not employ explanation strategies for service recovery, consumers are left unaware of the reasons for the failure, leading to uncertainty about responsibility attribution. The consumers will seek additional information to assist their next decision-making steps, at which point they will focus on information regarding the identity of the chatbot. Compared to chatbots with undisclosed identities, consumers have lower expectations of conversational competence for chatbots with

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disclosed identities, because they are already aware that the chatbot is not human (Grimes *et al.*, 2021). In addition, the act of proactive disclosure leads consumers to perceive chatbots as being sincere. After a service failure, consumers will then experience weaker negative emotions, because their expectations are lower, thereby effectively increasing their tolerance. Conversely, consumers may perceive non-disclosed (vs disclosed) chatbots as unconcerned with their feelings and not making ample efforts to resolve the problem, which will aggravate their disgust and perception of insincerity. The following hypothesis is thus proposed:

*H4a.* Chatbot explanation moderates the effect of identity disclosure on disgust, in which the absence of an explanation makes non-disclosure (vs disclosure) increase disgust, thereby lowering customer tolerance of service failure.

*H4b.* Chatbot explanation moderates the effect of identity disclosure on perceived sincerity, in which the absence of an explanation makes non-disclosure (vs disclosure) decrease perceived sincerity, thereby lowering customer tolerance of service failure.

Figure 1 shows the research model.

## 4. Method

### *4.1 Overview of studies*

The proposed hypotheses were tested through three scenario-based experiments (Table 2). To verify the robustness of the research results, typical scenarios (such as tourism taxes, beaches, insurance and science museums) were selected for study. Each scenario represents a distinct service domain, enabling a comprehensive assessment of chatbot interactions across various industries. Participants were recruited via the online survey platform Credamo.com and offered monetary compensation for their participation.

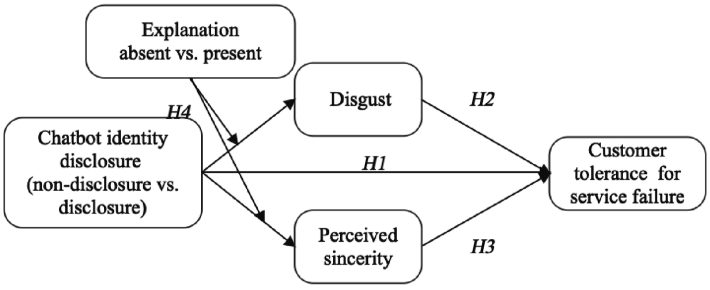



**Figure 1.** Research model. Source: Figure by authors

**Table 2.** Overview of studies

| Hypotheses | Studies | Designs | Scenarios |
|---|---|---|---|
| H1 | Study 1 | Chatbot identity disclosure (non-disclosure vs disclosure) | Tourist tax |
| H1, H2, H3 | Study 2 | Chatbot identity disclosure (non-disclosure vs disclosure) × Service context (hedonic vs utilitarian) | Beaches, Insurance |
| H1, H2, H3, H4a, and H4b | Study 3 | Chatbot identity disclosure (non-disclosure vs disclosure) × Explanation (absent vs present) | Science museum |

**Source(s):** Table by authors

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*4.2 Study 1*

Study 1 tested the effect of chatbot identity disclosure on customer tolerance of service failure.

*4.2.1 Pretest.* A total of 30 valid responses (43.3% male, $M_{age} = 30.47$, $SD = 8.57$) were collected from Credamo for the pretest. Participants were asked to read a scenario and imagine themselves as a customer visiting an online travel agency through their website, Xiaobai Trip. The online travel agency provided customers with travel advice, reservation services for plane tickets, hotel bookings and other forms of travel assistance. Participants imagined that they would be traveling to Bali next month, but they wanted to communicate with the agent first about some tourist tax issues. The chatbot was programmed to disclose that it was a robot during the self-introduction. After the introduction, the consumer engaged in an interaction with the chatbot. However, at the end of the interaction, the chatbot made a response error and failed to recognize the content of the consumer's question about the tourism tax (Appendix A).

Three items were used to measure participants' perceptions of chatbot identity disclosure (Cheng *et al.*, 2022; Luo *et al.*, 2019). All items were scored on a 7-point Likert scale from 1 (*strongly disagree*) to 7 (*strongly agree*) (Appendix B). The one-way ANOVA results validate the effectiveness of this experimental manipulation of users' perceptions of chatbot identity disclosure ($\alpha = 0.772$, $M_{\text{non-disclosure}} = 3.07$, $M_{\text{disclosure}} = 5.98$, $F(1, 28) = 75.79$, $p < 0.001$).

*4.2.2 Main study.* A total of 70 participants (25.7% male, $M_{age} = 29.59$, $SD = 8.85$) were recruited from Credamo. They were randomly assigned to one of the two scenarios used in the pretest (see above). Participants were asked to imagine themselves as the customer described in the scenario. After reading the descriptive texts, participants were required to complete a questionnaire. The measurement instrument for the manipulation check of chatbot identity disclosure was the same as that used in the pretest. Tolerance of service failure was measured using three items adapted from De Matos *et al.* (2009) and Lv *et al.* (2021), where all responses were scored on a seven-point Likert scale, from 1 (*not at all*) to 7 (*very much*). The severity of failure was also measured (Weun *et al.*, 2004). Demographic data were also collected from the participants, and the discriminant validity matrices are detailed in Appendix C.

*4.2.3 Results.* 4.2.3.1 Manipulation check. The results of the one-way ANOVA confirm the effective manipulation of identity disclosure ($\alpha = 0.836$, $M_{\text{non-disclosure}} = 3.29$, $M_{\text{disclosure}} = 6.25$, $F(1, 68) = 144.33$, $p < 0.001$). This result shows that the effect of service failure severity was not significant ($F(1.68) = 2.95$, $p > 0.05$).

4.2.3.2 Main analysis. Non-disclosure (vs disclosure) evoked lower customer tolerance of service failure ($\alpha = 0.795$), according to one-way ANOVA ($M_{\text{non-disclosure}} = 4.06$, $M_{\text{disclosure}} = 4.83$, $F(1.68) = 8.24$, $p < 0.05$); H1 was thus supported.

*4.2.4 Discussion.* Study 1 confirmed that chatbot identity disclosure (vs non-disclosure) promoted customer tolerance of service failure.

*4.3 Study 2*

Study 2 had three main goals. First, the mediating roles of disgust and perceived sincerity were examined. Second, several alternative mechanisms were tested, namely attribution, anger and contempt. Finally, although tourist behavior is essentially a pleasure-seeking and leisurely activity, some aspects of travel, such as tax refunds and foreign exchange rates, are more practical (Liu *et al.*, 2022). Therefore, Study 2 was set in insurance-related and beach service-related settings to exclude the influence of context.

*4.3.1 Pretest.* A pretest was conducted to determine the effectiveness of service context and identity disclosure. A total of 46 participants were recruited from Credamo (30.4% male, $M_{age} = 30$, $SD = 7.26$) and were asked to read a scenario and behave as a customer visiting the online travel agency website Xiaobai Trip. In the scenario, participants were planning to travel to Phuket Island in Thailand the following month, but they had some questions about the trip that they wanted to communicate to the agent. The same identity disclosure process was used as in Study 1. Following the initial introduction, the consumer proceeded to interact with the chatbot. However, at the conclusion of the interaction, the chatbot encountered a simulated

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malfunction and provided an incorrect response (Appendix A). The service contexts included one in which the participants consulted the agent about insurance problems (utilitarian-dominant) and one in which the traveler asked about beach-related problems (hedonic-dominant) (Liu *et al.*, 2022). Afterwards, participants were asked to evaluate service attributes (Voss *et al.*, 2003; $\alpha = 0.970$) and identity disclosure ($\alpha = 0.825$) using a Likert scale. The measures for the above constructs are provided in Appendix B.

The results show that those seeking service in a hedonistic context scored service attributes lower than those seeking service in a utilitarian context ($M_{\text{hedonic}} = -1.81$ vs $M_{\text{utilitarian}} = 2.12$, $F(1.44) = 131.85$, $p < 0.001$). The non-disclosure group scored identity disclosure lower than the disclosure group ($M_{\text{non-disclosure}} = 3.15$, $M_{\text{disclosure}} = 6.02$, $F(1, 44) = 76.80$, $p < 0.001$). Therefore, the manipulation was successful.

*4.3.2 Main study.* A total of 107 participants (male = 45.8%, $M_{\text{age}} = 25.36$, $SD = 4.92$) were recruited from Credamo to participate in the formal experiment of Study 2. Participants were randomly assigned to one of the conditions in a 2 (non-disclosure vs disclosure) × 2 (utilitarian vs hedonic) between-subjects design.

Participants were also randomly assigned to read one of the scenarios from the pretest. During the interaction, to ensure data quality, the participants had to answer task-related questions as an attention check. The service attribute and identity disclosure measures were the same as described above for the pretest. Disgust, perceived sincerity and tolerance of service failure were then measured. Since consumers' attribution of service failure could have repercussions on their attitudes, attribution was also measured ($\alpha = 0.891$) (Yalcin *et al.*, 2022). Studies have shown that when a negative stimulus originates from external sources, consumers experience three distinct negative emotions: disgust (focusing on avoidance), anger (focusing on punishment) and contempt (focusing on prejudice) (Westbrook, 1987). Thus, anger and contempt were also measured to rule out any possible effects from those emotions. Finally, the severity of service failure was measured, and demographic variables were gathered (Appendix B). Discriminant validity matrices are detailed in Appendix C.

*4.3.3 Results.* 4.3.3.1 Manipulation check. The findings of this study show that the identity disclosure and service attribute manipulations were successful. The participants who experienced disclosure rated identity disclosure higher than those who experienced non-disclosure ($\alpha = 0.898$, $M_{\text{disclosure}} = 6.25$, $M_{\text{non-disclosure}} = 3.44$, $F(1, 105) = 166.59$, $p < 0.001$). The score of service attributes in the hedonic context was lower than that in the utilitarian context ($\alpha = 0.934$, $M_{\text{hedonic}} = -1.41$, $M_{\text{utilitarian}} = 1.63$, $F(1, 105) = 116.43$, $p < 0.001$). The result proved that the effect of service failure severity was not significant ($F(1,105) = 1.57$, $p > 0.05$).

4.3.3.2 Main analysis. The chatbot that disclosed its identity evoked higher customer tolerance of service failure than the chatbot that did not disclose its identity ($\alpha = 0.755$), as shown by the one-way ANOVA ($M_{\text{disclosure}} = 4.25$, $M_{\text{non-disclosure}} = 3.75$, $F(1,105) = 3.98$, $p < 0.05$); H1 is thus supported. A one-way ANOVA indicated that the effects of identity disclosure on disgust ($\alpha = 0.917$, $F(1,105) = 5.74$, $p < 0.05$) and perceived sincerity ($\alpha = 0.954$, $F(1,105) = 5.83$, $p < 0.05$) were significant. The non-disclosure group scored disgust significantly higher than the disclosure group ($M_{\text{non-disclosure}} = 4.64$, $M_{\text{disclosure}} = 3.93$). Furthermore, the non-disclosure group scored perceived sincerity significantly lower than the disclosure group ($M_{\text{non-disclosure}} = 3.41$, $M_{\text{disclosure}} = 4.14$). The mediating effects of disgust and perceived sincerity on the influence of chatbot identity disclosure on customer tolerance of service failure were tested using PROCESS (Model 4; 5,000 samples). The total effect of chatbot identity disclosure on tolerance of service failure was substantial (Effect = 0.4922, SE = 0.2469, 95% CI = [0.0027.0.9817]); H1 is thus supported. Disgust (Effect = 0.1077, SE = 0.061, 95% CI = [0.0031.0.2411]) and perceived sincerity (Effect = 0.2110, SE = 0.0943, 95% CI = [0.0394.0.4103]) mediated the relationship between chatbot identity disclosure and tolerance of service failure. Furthermore, when the two mediators were controlled, the direct effect of identity disclosure on tolerance of service failure was found to be insignificant (Effect = 0.0797,

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SE = 0.2048, 95% CI = [−0.3264.0.4858]), proving the essential mediating roles of disgust and perceived sincerity. Therefore, H2 and H3 are supported.

ANOVA was used to rule out interference from service attributes. The influence of service attributes on customer tolerance of service failure was found to be insignificant ($M_{\text{hedonic}} = 3.95$, $M_{\text{utilitarian}} = 4.04$, $F$ (1, 105) = 0.14, $p > 0.05$). Moreover, the two-way interaction between identity disclosure and service attributes similarly did not have a significant effect ($F$ (3, 103) = 0.19, $p > 0.05$). This finding effectively eliminates any potential impact of service attributes on our results.

Alternative mechanisms were examined as well. The first to be assessed was attribution. ANOVA revealed that the effect of identity disclosure on attribution was not significant ($M_{\text{disclosure}} = 2.95$, $M_{\text{non-disclosure}} = 2.78$, $F$ (1, 105) = 0.38, $p > 0.05$), precluding identity disclosure as a driver for the observed mediation pattern. Anger and contempt were then assessed as possible alternative mechanisms. The main effects of identity disclosure on anger ($F$ (1, 105) = 3.08, $p > 0.05$) and on contempt ($F$ (1, 105) = 3.25, $p > 0.05$) were not significant. Together, these results exclude many alternative mechanisms.

*4.3.4 Discussion.* Study 2 found that chatbot identity disclosure (vs non-disclosure) positively affected customer tolerance of service failure. The mediating roles of disgust and perceived sincerity in the impact of identity disclosure were also verified. In addition, this study excluded the influence of service context. Finally, multiple alternative explanations for this study were ruled out, including attribution, anger and contempt.

### *4.4 Study 3*

Study 3 investigated the possible moderating role of the presence of explanation in the relationship between identity disclosure and consumer tolerance.

*4.4.1 Participants and design.* For Study 3, 196 individuals (male = 48.5%, $M_{\text{age}} = 32.85$, $SD = 9.08$) were recruited from Credamo. Participants were randomly assigned to one of the conditions in a 2 (non-disclosure vs disclosure) × 2 (no explanation vs explanation) between-subjects design.

The participants were asked to imagine planning a visit to their local science museum over the weekend. To gain advanced knowledge of any special exhibitions at the museum, they decided to consult online customer service. Subsequently, they opened the official mini-program of the science museum and established contact with the online customer service chatbot. After the chatbot introduced itself in a manner identical to that used in Study 1, the participants were shown a picture of a failed booking during the chat process. In the no-explanation group, the chatbot did not offer any explanation for the fault and merely indicated the hope that the consumer would try the interaction again at another time. In the explanation group, the chatbot provided an explanation for the communication failure by stating that it was due to an unstable power system causing network delays. In that case, the chatbot expressed the hope that the consumer would attempt the interaction at a different time (Appendix A). Subsequently, the participants evaluated their perceptions of disclosure ($\alpha = 0.853$), disgust ($\alpha = 0.905$), perceived sincerity ($\alpha = 0.928$) and tolerance ($\alpha = 0.851$) using a Likert scale. The details of the evaluation measures are provided in Appendix B. Finally, participants rated the severity of service failure and provided their demographic information. Discriminant validity matrices are detailed in Appendix C.

*4.4.2 Results.* 4.4.2.1 Manipulation check. The results of one-way ANOVA verified the effective manipulation of identity disclosure ($M_{\text{non-disclosure}} = 4.09$, $M_{\text{disclosure}} = 6.08$, $F$ (1,194) = 102.12, $p < 0.001$). This result proves that the effect of service failure severity was not significant ($F$ (1,194) = 1.33, $p > 0.05$).

4.4.2.2 Main analysis. One-way ANOVA indicated that disclosure (vs non-disclosure) evoked greater consumer tolerance ($M_{\text{disclosure}} = 4.82$, $M_{\text{non-disclosure}} = 4.38$, $F$ (1,194) = 4.98, $p < 0.05$); H1 is thus supported. Importantly, the interaction of identity disclosure × explanation was found to have a significant effect on consumer tolerance ($F$ (3,

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194) = 4.04, $p < 0.05$). In the no-explanation scenario, participants were more tolerant when the chatbot identity was disclosed than when it was not ($M_{\text{non-disclosure}} = 3.48$, $M_{\text{disclosure}} = 4.18$, $F(1, 90) = 7.42$, $p < 0.05$). For those in the explanation scenario, disclosure had no significant effect on consumer tolerance ($M_{\text{non-disclosure}} = 5.27$, $M_{\text{disclosure}} = 5.29$, $F(1,104) = 0.01$, $p > 0.05$). The results are presented in Figure 2.

Furthermore, one-way ANOVA demonstrated that identity disclosure had a significant impact on both disgust ($F(1,194) = 4.63$, $p < 0.05$) and perceived sincerity ($F(1,194) = 4.17$, $p < 0.05$). The non-disclosure group rated their disgust significantly higher than the disclosure group ($M_{\text{non-disclosure}} = 3.58$, $M_{\text{disclosure}} = 3.11$). In addition, the non-disclosure group rated perceived sincerity significantly lower than the disclosure group ($M_{\text{non-disclosure}} = 4.41$, $M_{\text{disclosure}} = 4.82$). Two-way ANOVA indicated that the effects of the interaction between identity disclosure and explanation on both disgust ($F(3,193) = 4.59$, $p < 0.05$) and perceived sincerity ($F(3,193) = 5.57$, $p < 0.05$) were significant. In the no-explanation condition, non-disclosure generated more disgust ($M_{\text{non-disclosure}} = 4.56$, $M_{\text{disclosure}} = 3.79$, $F(1.90) = 8.32$, $p < 0.05$) and less perceived sincerity ($M_{\text{non-disclosure}} = 3.47$, $M_{\text{disclosure}} = 4.19$, $F(1.90) = 6.94$, $p < 0.05$) than non-disclosure did (Figures 3 and 4). In the explanation condition, disclosure had no significant difference in effect on disgust ($F(1,104) = 0.005$, $p > 0.05$) and no significant effect on perceived sincerity ($F(1,104) = 0.12$, $p > 0.05$), supporting H4a and H4b.

Finally, a moderation analysis in PROCESS, Model 7 (5,000 samples) was used to test whether offering an explanation influenced the relationships among identity disclosure, disgust, perceived sincerity and tolerance. Disgust (Effect = −0.1809, SE = 0.1068, 95% CI = [−0.4126, −0.0065]) and perceived sincerity (Effect = −0.4975, SE = 0.2263, 95% CI = [−0.9612, −0.0745]) were both found to mediate the relationships linking tolerance to both disclosure and explanation, proving that both disgust and perceived sincerity play mediating roles (Table 3).

*4.4.3 Discussion.* Study 3 confirmed that identity disclosure and explanation affect consumer tolerance, disgust and perceived sincerity. In the no-explanation condition, the effect of disclosure on disgust was attenuated, while the effect of disclosure on perceived sincerity was enhanced, thereby increasing consumer tolerance. When an explanation was given, the effects of disclosure on disgust, perceived sincerity, and consumer tolerance did not change significantly.

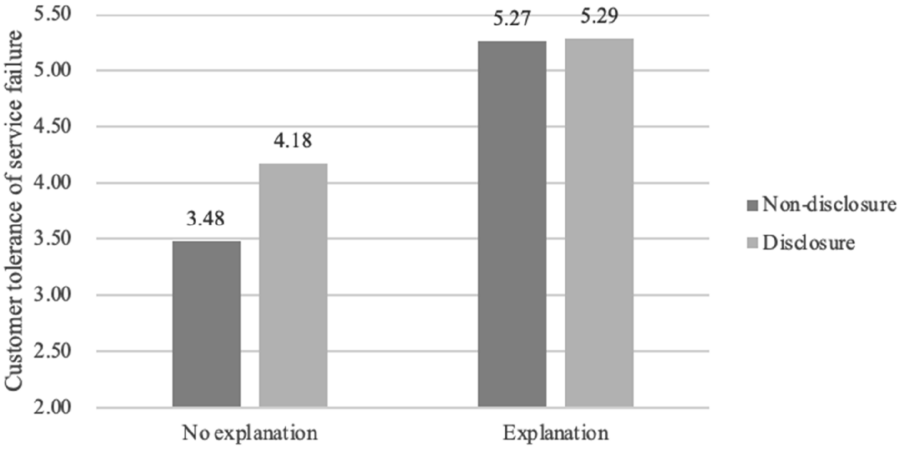



**Figure 2.** Impact of the interaction between identity disclosure and explanation on customer tolerance of service failure. Source: Figure by authors

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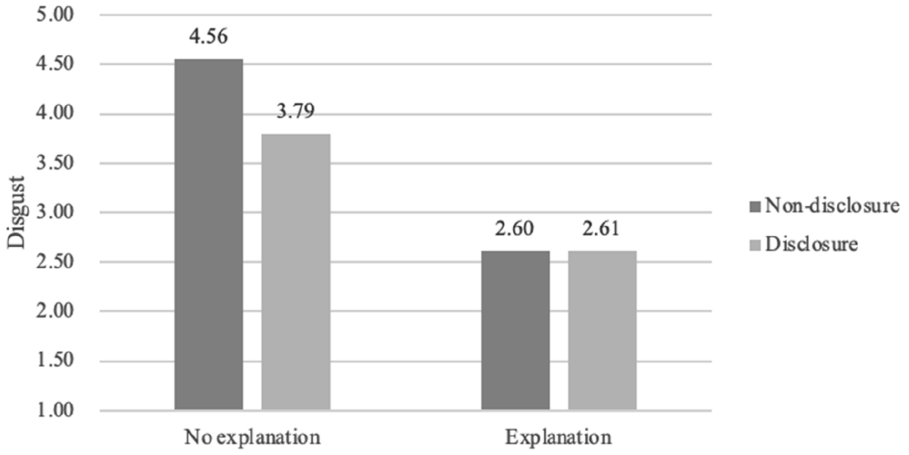



**Figure 3.** Impact of the interaction between disclosure and explanation on disgust. Source: Figure by authors

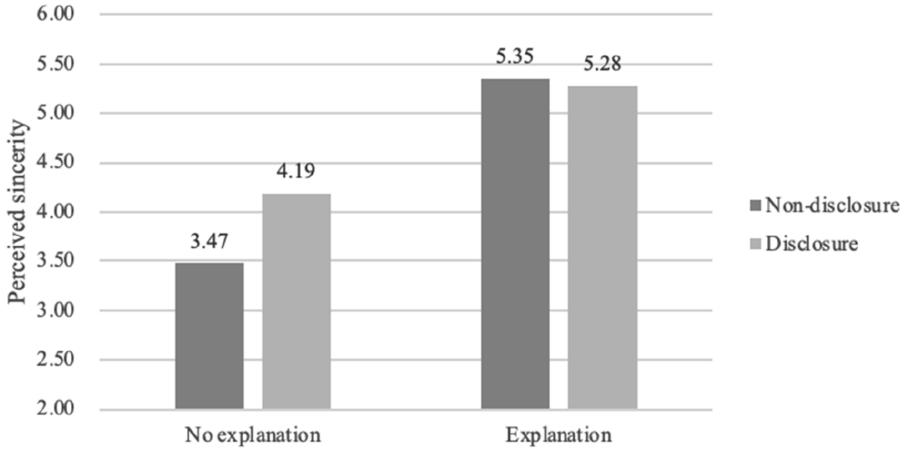



**Figure 4.** Impact of the interaction between disclosure and explanation on perceived sincerity. Source: Figure by authors

## 5. Conclusions

The use of chatbots to provide customer service is becoming ever more widespread in the service industries. However, chatbots' technical limitations mean they are prone to service failure (Crolic *et al.*, 2022). Although some customers enjoy interacting with chatbots, others dislike them, especially when such interactions make them feel as if they have been cheated (Beeler *et al.*, 2022). Therefore, the focus of this study is to understand the role of identity disclosure by chatbots in shaping consumer responses during service failures.

Within this context, this study focuses specifically on exploring the impacts of chatbot identity disclosure on consumer tolerance of service failure, as well as the internal mechanisms involved. Study 1 shows that consumers were more likely to tolerate service failures from disclosed (vs non-disclosed) chatbots. Study 2 confirms that disgust and perceived sincerity mediate the relationship between chatbot identity disclosure and customer tolerance of service

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**Table 3.** Moderated mediation analysis

| Effect | Path | Moderator | Effect | SE | 95% confidence interval LLCI | ULCI |
|---|---|---|---|---|---|---|
| Direct effect | Disclosure × Explanation → Consumer tolerance of service failure | – | 0.0792 | 0.113 | −0.1436 | 0.3021 |
| Indirect effect | Disclosure × Explanation → Disgust → Consumer tolerance of service failure | – | −0.1809 | 0.1068 | −0.4126 | −0.0065 |
| | | No explanation | 0.177 | 0.0886 | 0.0352 | 0.3733 |
| | | Explanation | −0.004 | 0.0606 | −0.1276 | 0.124 |
| | Disclosure × Explanation → Perceived sincerity → Consumer tolerance of service failure | – | −0.4975 | 0.2263 | −0.9612 | −0.0745 |
| | | No explanation | 0.4534 | 0.1864 | 0.1049 | 0.8299 |
| | | Explanation | −0.0441 | 0.1274 | −0.292 | 0.2102 |

**Source(s):** Table by authors

failure. Meanwhile, the impact of service contexts and other potential mechanisms was excluded, including the mechanisms of attribution, anger and contempt. Study 3 reveals that when a chatbot does not explain the reasons for service failure, disclosure (vs non-disclosure) elicits lower levels of disgust and higher levels of perceived sincerity, thus increasing customer tolerance of service failure. When explaining the reasons for service failure, disclosure (vs non-disclosure) does not significantly change disgust, perceptions of sincerity or customer tolerance of service failure.

### *5.1 Theoretical implications*

This study makes a number of contributions to the relevant literature. First, the study enriches the academic research on identity disclosure by chatbots in the context of service failure. Prior studies have predominantly focused on recovery measures, such as appearance design, language content and linguistic style (Choi *et al.*, 2021; Lv *et al.*, 2022; Song *et al.*, 2023; Xu and Liu, 2022), while overlooking the role of identity information. In addition, although some studies have explored the negative impacts of identity disclosure (De Cicco *et al.*, 2021; Li *et al.*, 2024; Luo *et al.*, 2019; Powers *et al.*, 2023), studies on the positive effects of identity disclosure are comparatively scarce (Beeler *et al.*, 2022; Kim, 2025), particularly in the context of service failures. Mozafari *et al.* (2022) found that chatbot identity disclosure after a service failure can enhance customer trust and retention. However, offering alternative service options in the later stages of service failure may lead consumers to adopt emotional coping strategies and potentially even attack customer service representatives (Huang and Dootson, 2022). The present study instead focuses on the initial phase – before service failure – and examines how the disclosure of a chatbot's identity influences consumers' tolerance of service failures, thereby addressing a gap in existing literature.

Second, this work reveals the mediating effects of disgust and perceived sincerity on the link between chatbot identity disclosure and customer tolerance of service failure. Prior studies on tolerance of chatbot service failure have considered positive aspects, such as tenderness, performance expectancy (Lv *et al.*, 2021) and perceived competence and warmth (Xu and Liu, 2022). However, researchers have rarely examined changes in customers' negative emotions and perceptions of sincerity toward agents. This paper focuses on how consumers' service failure tolerance is affected by identity disclosure in terms of their perceptions of chatbots and the changes in their own emotions. The double mediation of disgust and perceived sincerity is revealed through situational experiments; this paper also adds new content to the study of the mechanisms of influences on customers' tolerance of service failure.

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Third, this study investigates the moderating role of offered explanations in the impact of identity disclosure on customer tolerance of service failure, thus clarifying the boundary conditions of that influence. In recent years, the economic and social significance of explainable AI has captured the interest of researchers (Arrieta *et al.*, 2020). The goal of providing explanations is to enhance transparency, allowing users to comprehend the technical context and decision-making procedures involved in their interactions with AI systems (Akalin *et al.*, 2022; Lee *et al.*, 2023). The findings of this paper support the idea that explaining how algorithms operate can shape consumers' attitudes toward those algorithms (Ostinelli *et al.*, 2025; Yeomans *et al.*, 2019). This not only deepens the understanding of the function of explanations but also offers theoretical backing for the formulation of service recovery strategies.

### *5.2 Practical implications*

This study has valuable practical implications for service companies. First, companies should seriously consider the question of chatbot identity disclosure. As chatbots gradually replace human employees, it is important to consider customers' views on disclosure (Beeler *et al.*, 2022). It is not advisable to conceal the identity of a chatbot. For example, Google has received criticism for the search engine's lack of transparency in regard to AI interactions with customers. To enhance post-service failure tolerance, companies should consider revealing the true identity of chatbots via communication interface designs, such as text-based introductions or avatar displays.

Second, chatbot service failure is inevitable. Therefore, enterprises should provide information about the chatbot prior to the chatbot's interactions with consumers, to decrease the possibility of negative consequences in case of service failure and to manage customers' expectations. The lower their expectations are, the weaker the negative reactions triggered by service failure will be. Thus, chatbot identity disclosure prior to interactions can alleviate consumers' disgust and dissatisfaction and shape their expectations of potential service failure. For example, companies could use language or other means to help customers better understand chatbots, Doing so may help prevent customers from reacting to the failure of chatbot services with negative emotions and behavior.

Third, companies should decide whether to reveal the true identity of a chatbot based on their own capability to provide service recovery. If a company cannot offer targeted explanations for the service issues consumers encounter, disclosing the chatbot's non-human identity in advance can help mitigate the negative impacts of service failure. Conversely, if a company has the capacity to provide reasonable explanations and other service recovery tactics, the question of whether the chatbot discloses its identity becomes less critical.

### *5.3 Limitations and future research*

This article has some limitations that point to directions for future research. First, this study only considers text-based chatbots. Subsequent studies should investigate whether the same effects occur with voice chatbots. Second, this study only considers language manipulation. In the future, avatar manipulation could be considered as well. Third, in addition to explanation, other situational variables such as time pressure and consumers' initial emotions may moderate the disclosure effect. Therefore, these variables should be considered in follow-up studies. Finally, this study only uses scenario-based experiments to verify the hypotheses. Future work should consider field experiments in actual interaction contexts to increase the external validity of the research results.

**Supplementary material**

The supplementary material for this article can be found online.

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